Arijit Ghosh

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Education

M.Sc. in Artificial Intelligence **Oct 2021 - Sep 2024**

Friedrich Alexander University, Erlangen–Nürnberg

Current Grade: **1.3** (Best: 1.0, Worst: 4.0)
Minor: Artificial Intelligence in Biomedical Engineering
Master Thesis: Exploring Disentagled Structure-Texture Generative Models on Image-To-Image Translation Task.

B. Tech. in Electronics and Communication Engineering **Aug 2017 - Jul 2021**

Maulana Abul Kalam University of Technology (formerly West Bengal University of Technology)

Final Grade: **9.24** (Best: 10.0, Worst: 5.0)

Research Experience

Student Teaching and Research Assistant **Jan 2022 - Present**

IDEA Lab, FAU, Erlangen–Nürnberg (Supervisor: **Prof. Dr. Bernhard Kainz**)

Student Research Assistant **Mar 2023 - Present**

Fraunhofer IIS, Nürnberg (Supervisor: **Mark Deutel, M.Sc.**)

Student Research Intern **Apr 2020 - Sep 2021**

Maulana Abul Kalam University of Technology (Supervisor: **Dr. Soumen Pati)**

Work Experience

Working Student - Deep Learning and Computer Vision **Sep 2022 - Feb 2023**

Primetals Technologies, Erlangen (Supervisor: **Dominik Wassermann, M.Sc.**)

Mentor - Generative Adversarial Networks Specialization **Aug 2021 - Dec 2021**

Deeplearning.AI

Publications

Roy, I., Shai, R., **Ghosh, A.**, Bej, A. and Pati, S.K., 2021. CoWarriorNet: A novel deep-learning framework for COVID-19 detection from chest X-ray images. New Generation Computing, pp.1-25.

Pati, S.K., **Ghosh, A.**, Banerjee, A., Roy, I., Ghosh, P. and Kakar, C., 2021. Data analysis on cancer disease using machine learning techniques. In Advanced Machine Learning Approaches in Cancer Prognosis: Challenges and Applications, pp. 13-73.

Bej, A., Roy, I., Chanda, S., **Ghosh, A.** and Pati, S.K., 2022. Feature Mining and Classification of Microarray Data Using Modified ResNet-SVM Architecture. In Computational Intelligence in Pattern Recognition: Proceedings of CIPR 2021, pp. 317-328.

Pati, S.K., Gupta, M.K., Shai, R., Banerjee, A. and **Ghosh, A.**, 2022. Missing value estimation of microarray data using Sim-GAN. Knowledge and Information Systems, 64(10), pp.2661-2687.

Ghosh, A., Roy I., Gupta, M.K., Banerjee, A. and Pati, S.K., 2022. Deep Learning based Gene Selection on Microarray Data: A Review. *Under Review.*

Ghosh, A., Dombrowski, M., Reynaud, H. and Kainz, B., 2023. DESTINE: A Fresh Look at Disentanglement based Single-Image Image-to-Image Translation. *Under Review.*.

Technical and Specialized Skills

Languages/Frameworks: Confident with Pytorch, Pytorch-Lightning, Monai, Python, Git.

Fields of Interest: Deep Learning, Machine Learning, Medical Image Analysis, Image-to-Image Translation, Representation Learning, Self-Supervised Learning, Contrastive Learning, Semi-Supervised Learning, Weakly-Supervised Learning.

Projects

Master's Project-1 : Segmentation with a punch via VoxelMorph and Registration.
Supervisor: **Prof. Dr. Bernhard Kainz**

Master's Project-2 : Fake it till you Can't make it.
Supervisor: **Prof. Dr. Bernhard Egger**

Achievements

Winner of Researchathon 2021 in Healthcare Track **July 2021**

Referees

Prof. Dr. Bernhard Kainz : bernhard.kainz@fau.de
Prof. Dr. Bernhard Egger : bernhard.egger@fau.de
Mr. Mark Deutel, M.Sc.: mark.deutel@iis-extern.fraunhofer.de